

Lecture 3: Intro to proofs for algorithms

Jessica Sorrell

September 8, 2025
601.433/633 Introduction to Algorithms

Today

Master theorem and common proof techniques for algorithms.

- ▶ Inductive arguments (weak, strong)
- ▶ Proof by contradiction
- ▶ Direct proof
- ▶ Loop invariant
- ▶ Proof by contrapositive

We'll demonstrate proof techniques by proving the correctness and running time of algorithms you've seen before.

Quicksort review

Algorithm Quicksort

Input: array $\mathbf{A}$ of length n

- 1: **if** $n \leq 1$ **then**
 - 2: **return** $\mathbf{A}$
 - 3: **end if**
 - 4: Pick some element $p \in \mathbf{A}$ as the *pivot*
 - 5: Let $\mathbf{L}$ be the elements less than or equal to p , let $\mathbf{G}$ be the elements larger than p
 - 6: $\mathbf{L}' \leftarrow \text{Quicksort}(\mathbf{L})$
 - 7: $\mathbf{G}' \leftarrow \text{Quicksort}(\mathbf{G})$
 - 8: **return** $\mathbf{L}' \parallel p \parallel \mathbf{G}'$
-

Strong Induction vs Weak Induction

Strong induction:

- ▶ Prove property holds for a base case
- ▶ e.g. Quicksort always returns a sorted array for input arrays of size $n \leq 1$

Strong Induction vs Weak Induction

Strong induction:

- ▶ Prove property holds for a base case
- ▶ e.g. Quicksort always returns a sorted array for input arrays of size $n \leq 1$
- ▶ Assume inductive hypothesis, that property holds *for all* $n \leq k$. Then show that property holds for $n = k + 1$.
- ▶ e.g. Assume Quicksort always returns a sorted array for input arrays of size $\leq k$. Show it returns a sorted array for input arrays of size $k + 1$.

Strong Induction vs Weak Induction

Strong induction:

- ▶ Prove property holds for a base case
- ▶ e.g. Quicksort always returns a sorted array for input arrays of size $n \leq 1$
- ▶ Assume inductive hypothesis, that property holds *for all* $n \leq k$. Then show that property holds for $n = k + 1$.
- ▶ e.g. Assume Quicksort always returns a sorted array for input arrays of size $\leq k$. Show it returns a sorted array for input arrays of size $k + 1$.

Weak induction:

- ▶ Prove property holds for a base case

Strong Induction vs Weak Induction

Strong induction:

- ▶ Prove property holds for a base case
- ▶ e.g. Quicksort always returns a sorted array for input arrays of size $n \leq 1$
- ▶ Assume inductive hypothesis, that property holds *for all* $n \leq k$. Then show that property holds for $n = k + 1$.
- ▶ e.g. Assume Quicksort always returns a sorted array for input arrays of size $\leq k$. Show it returns a sorted array for input arrays of size $k + 1$.

Weak induction:

- ▶ Prove property holds for a base case
- ▶ Assume inductive hypothesis, that property holds for $n = k$. Then show that property holds for $n = k + 1$.
- ▶ e.g. Assume Quicksort always returns a sorted array for input arrays of size *exactly* k . Show it returns a sorted array for input arrays of size $k + 1$.

Correctness of Quicksort - (Strong) Inductive Proof

Claim: Given an array $\mathbf{A}$ of length n , Quicksort($\mathbf{A}$) returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Correctness of Quicksort - (Strong) Inductive Proof

Claim: Given an array $\mathbf{A}$ of length n , Quicksort($\mathbf{A}$) returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Proof:

- ▶ Base case: $n \leq 1$. Quicksort($\mathbf{A}$) returns $\mathbf{A}$.

Correctness of Quicksort - (Strong) Inductive Proof

Claim: Given an array $\mathbf{A}$ of length n , Quicksort($\mathbf{A}$) returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Proof:

- ▶ Base case: $n \leq 1$. Quicksort($\mathbf{A}$) returns $\mathbf{A}$.
- ▶ Inductive step: Assume Quicksort($\mathbf{A}$) returns a sorted array for all $\mathbf{A}$ of length $\leq n$. Show it returns a sorted array for all $\mathbf{A}$ of length $n + 1$.

Correctness of Quicksort - (Strong) Inductive Proof

Claim: Given an array $\mathbf{A}$ of length n , Quicksort($\mathbf{A}$) returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Proof:

- ▶ Base case: $n \leq 1$. Quicksort($\mathbf{A}$) returns $\mathbf{A}$.
- ▶ Inductive step: Assume Quicksort($\mathbf{A}$) returns a sorted array for all $\mathbf{A}$ of length $\leq n$. Show it returns a sorted array for all $\mathbf{A}$ of length $n + 1$.
 - ▶ Pick pivot $p \in \mathbf{A}$. Let $\mathbf{L}$ be the elements less than or equal to p , let $\mathbf{G}$ be the elements larger than p .

Correctness of Quicksort - (Strong) Inductive Proof

Claim: Given an array $\mathbf{A}$ of length n , Quicksort($\mathbf{A}$) returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Proof:

- ▶ Base case: $n \leq 1$. Quicksort($\mathbf{A}$) returns $\mathbf{A}$.
- ▶ Inductive step: Assume Quicksort($\mathbf{A}$) returns a sorted array for all $\mathbf{A}$ of length $\leq n$. Show it returns a sorted array for all $\mathbf{A}$ of length $n + 1$.
 - ▶ Pick pivot $p \in \mathbf{A}$. Let $\mathbf{L}$ be the elements less than or equal to p , let $\mathbf{G}$ be the elements larger than p .
 - ▶ $\mathbf{L}$ and $\mathbf{G}$ are of length $\leq n$, so by inductive hypothesis, Quicksort($\mathbf{L}$) and Quicksort($\mathbf{G}$) return sorted arrays $\mathbf{L}'$ and $\mathbf{G}'$.

Correctness of Quicksort - (Strong) Inductive Proof

Claim: Given an array $\mathbf{A}$ of length n , Quicksort($\mathbf{A}$) returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Proof:

- ▶ Base case: $n \leq 1$. Quicksort($\mathbf{A}$) returns $\mathbf{A}$.
- ▶ Inductive step: Assume Quicksort($\mathbf{A}$) returns a sorted array for all $\mathbf{A}$ of length $\leq n$. Show it returns a sorted array for all $\mathbf{A}$ of length $n + 1$.
 - ▶ Pick pivot $p \in \mathbf{A}$. Let $\mathbf{L}$ be the elements less than or equal to p , let $\mathbf{G}$ be the elements larger than p .
 - ▶ $\mathbf{L}$ and $\mathbf{G}$ are of length $\leq n$, so by inductive hypothesis, Quicksort($\mathbf{L}$) and Quicksort($\mathbf{G}$) return sorted arrays $\mathbf{L}'$ and $\mathbf{G}'$.
 - ▶ Therefore $\mathbf{L}' \parallel p \parallel \mathbf{G}'$ is sorted.

Why Strong Induction?

- ▶ A weak inductive hypothesis assumes the desired property holds for $n = k$.

Why Strong Induction?

- ▶ A weak inductive hypothesis assumes the desired property holds for $n = k$.
- ▶ A strong inductive hypothesis assumes the desired property holds for all $n \leq k$.

Why Strong Induction?

- ▶ A weak inductive hypothesis assumes the desired property holds for $n = k$.
- ▶ A strong inductive hypothesis assumes the desired property holds for all $n \leq k$.
- ▶ Quicksort recursively calls itself on L and G , which we don't know the size of a priori
- ▶ In strong induction, we assume that Quicksort is correct for all arrays of size $\leq k$, so doesn't matter what the exact size L and G are, because we know they are both $\leq k$.

Correctness of Quicksort - Proof by Contradiction

Claim: Given an array $\mathbf{A}$ of length n , Quicksort($\mathbf{A}$) returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Correctness of Quicksort - Proof by Contradiction

Claim: Given an array $\mathbf{A}$ of length n , Quicksort($\mathbf{A}$) returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Proof:

- ▶ Assume there exists at least one array such that Quicksort does not return a sorted array. Let $\mathbf{A}$ be the smallest such array and let n be the size of $\mathbf{A}$.

Correctness of Quicksort - Proof by Contradiction

Claim: Given an array $\mathbf{A}$ of length n , Quicksort($\mathbf{A}$) returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Proof:

- ▶ Assume there exists at least one array such that Quicksort does not return a sorted array. Let $\mathbf{A}$ be the smallest such array and let n be the size of $\mathbf{A}$.
- ▶ Note $n \geq 2$.

Correctness of Quicksort - Proof by Contradiction

Claim: Given an array $\mathbf{A}$ of length n , Quicksort($\mathbf{A}$) returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Proof:

- ▶ Assume there exists at least one array such that Quicksort does not return a sorted array. Let $\mathbf{A}$ be the smallest such array and let n be the size of $\mathbf{A}$.
- ▶ Note $n \geq 2$.
- ▶ So Quicksort($\mathbf{A}$) picks a pivot element $p \in \mathbf{A}$, defines $\mathbf{L}$ and $\mathbf{G}$ as the elements less than or equal to p and the elements greater than p respectively, and recursively calls Quicksort on $\mathbf{L}$ and $\mathbf{G}$.

Correctness of Quicksort - Proof by Contradiction

Claim: Given an array $\mathbf{A}$ of length n , Quicksort($\mathbf{A}$) returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Proof:

- ▶ Assume there exists at least one array such that Quicksort does not return a sorted array. Let $\mathbf{A}$ be the smallest such array and let n be the size of $\mathbf{A}$.
- ▶ Note $n \geq 2$.
- ▶ So Quicksort($\mathbf{A}$) picks a pivot element $p \in \mathbf{A}$, defines $\mathbf{L}$ and $\mathbf{G}$ as the elements less than or equal to p and the elements greater than p respectively, and recursively calls Quicksort on $\mathbf{L}$ and $\mathbf{G}$.
- ▶ By assumption that $\mathbf{A}$ is the smallest such array, $\mathbf{L}$ and $\mathbf{G}$ are sorted.

Correctness of Quicksort - Proof by Contradiction

Claim: Given an array $\mathbf{A}$ of length n , Quicksort($\mathbf{A}$) returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Proof:

- ▶ Assume there exists at least one array such that Quicksort does not return a sorted array. Let $\mathbf{A}$ be the smallest such array and let n be the size of $\mathbf{A}$.
- ▶ Note $n \geq 2$.
- ▶ So Quicksort($\mathbf{A}$) picks a pivot element $p \in \mathbf{A}$, defines $\mathbf{L}$ and $\mathbf{G}$ as the elements less than or equal to p and the elements greater than p respectively, and recursively calls Quicksort on $\mathbf{L}$ and $\mathbf{G}$.
- ▶ By assumption that $\mathbf{A}$ is the smallest such array, $\mathbf{L}$ and $\mathbf{G}$ are sorted.
- ▶ Therefore $\mathbf{L} \parallel p \parallel \mathbf{G}$ is sorted.
- ▶ Contradiction: $\mathbf{A}$ is not the smallest array such that Quicksort does not return a sorted array.

Direct Proof

A direct proof argues the conclusion of a claim *directly* from its assumptions.

Direct Proof

A direct proof argues the conclusion of a claim *directly* from its assumptions.

For a statement of the form $\mathbf{A} \Rightarrow \mathbf{B}$, a direct proof shows that $\mathbf{B}$ follows from the logical implications of $\mathbf{A}$.

Quicksort Runtime - Direct Proof

Claim: Quicksort runs in time $O(n^2)$ in the worst case.

Quicksort Runtime - Direct Proof

Claim: Quicksort runs in time $O(n^2)$ in the worst case.

- ▶ Before making its two recursive calls, Quicksort compares every element of its input array to the pivot, taking time $\Theta(n)$.

Quicksort Runtime - Direct Proof

Claim: Quicksort runs in time $O(n^2)$ in the worst case.

- ▶ Before making its two recursive calls, Quicksort compares every element of its input array to the pivot, taking time $\Theta(n)$.
- ▶ The worst case for runtime occurs when the pivot is the smallest or largest element of the array.

Quicksort Runtime - Direct Proof

Claim: Quicksort runs in time $O(n^2)$ in the worst case.

- ▶ Before making its two recursive calls, Quicksort compares every element of its input array to the pivot, taking time $\Theta(n)$.
- ▶ The worst case for runtime occurs when the pivot is the smallest or largest element of the array.
- ▶ In this case, the array is partitioned into an array of size $n - 1$ and an array of size 0 .

Quicksort Runtime - Direct Proof

Claim: Quicksort runs in time $O(n^2)$ in the worst case.

- ▶ Before making its two recursive calls, Quicksort compares every element of its input array to the pivot, taking time $\Theta(n)$.
- ▶ The worst case for runtime occurs when the pivot is the smallest or largest element of the array.
- ▶ In this case, the array is partitioned into an array of size $n - 1$ and an array of size 0 .
- ▶ This gives a recurrence $T(n) = T(n - 1) + \Theta(n)$, which has solution $T(n) = \Theta(n^2)$.

Insertion Sort Review

Algorithm Insertion Sort

Input: array A of length n

```
1: for  $i \leftarrow 2$  to  $n$  do
2:    $j \leftarrow i$ 
3:   while  $j > 1$  and  $A[j] < A[j - 1]$  do
4:     Swap  $A[j]$  and  $A[j - 1]$ 
5:      $j \leftarrow j - 1$ 
6:   end while
7: end for
```

Proof by Loop Invariant

Proof by loop invariant is a proof technique that establishes some useful property that is true throughout every loop of an iterative algorithm.

Proof by Loop Invariant

Proof by loop invariant is a proof technique that establishes some useful property that is true throughout every loop of an iterative algorithm.

- ▶ Initialization: the property is true at the start of the loop.

Proof by Loop Invariant

Proof by loop invariant is a proof technique that establishes some useful property that is true throughout every loop of an iterative algorithm.

- ▶ Initialization: the property is true at the start of the loop.
- ▶ Maintenance: if the property is true at the beginning of an iteration, it is true at beginning of the next iteration.

Proof by Loop Invariant

Proof by loop invariant is a proof technique that establishes some useful property that is true throughout every loop of an iterative algorithm.

- ▶ Initialization: the property is true at the start of the loop.
- ▶ Maintenance: if the property is true at the beginning of an iteration, it is true at beginning of the next iteration.
- ▶ Termination: when the loop terminates, the invariant holds and shows that the algorithm is correct.

Correctness of Insertion Sort - Proof by Loop Invariant

Claim: Given an array $\mathbf{A}$ of length n , $\text{InsertionSort}(\mathbf{A})$ returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Correctness of Insertion Sort - Proof by Loop Invariant

Claim: Given an array $\mathbf{A}$ of length n , $\text{InsertionSort}(\mathbf{A})$ returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Proof:

- ▶ Loop invariant: at iteration i , $\mathbf{A}[1, i - 1]$ contains all elements of the original input array $\mathbf{A}[1, i - 1]$, and is sorted.

Correctness of Insertion Sort - Proof by Loop Invariant

Claim: Given an array $\mathbf{A}$ of length n , $\text{InsertionSort}(\mathbf{A})$ returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Proof:

- ▶ Loop invariant: at iteration i , $\mathbf{A}[1, i - 1]$ contains all elements of the original input array $\mathbf{A}[1, i - 1]$, and is sorted.
- ▶ Initialization - At the beginning of the first iteration $i = 2$, $\mathbf{A}[1]$ is sorted.
- ▶ Maintenance - In a single iteration, element $\mathbf{A}[i]$ of the input Array is moved to the left until it is no longer smaller than the element to its left, therefore at the beginning of the next iteration, $\mathbf{A}[1, i]$ is sorted and contains exactly the same elements as $\mathbf{A}[1, i]$ from the original input array.

Correctness of Insertion Sort - Proof by Loop Invariant

Claim: Given an array $\mathbf{A}$ of length n , $\text{InsertionSort}(\mathbf{A})$ returns an array with all elements of $\mathbf{A}$ sorted from least to greatest.

Proof:

- ▶ Loop invariant: at iteration i , $\mathbf{A}[1, i - 1]$ contains all elements of the original input array $\mathbf{A}[1, i - 1]$, and is sorted.
- ▶ Initialization - At the beginning of the first iteration $i = 2$, $\mathbf{A}[1]$ is sorted.
- ▶ Maintenance - In a single iteration, element $\mathbf{A}[i]$ of the input Array is moved to the left until it is no longer smaller than the element to its left, therefore at the beginning of the next iteration, $\mathbf{A}[1, i]$ is sorted and contains exactly the same elements as $\mathbf{A}[1, i]$ from the original input array.
- ▶ Termination - When the loop terminates, $i = n$ and therefore $\mathbf{A}[1, n]$ is sorted and contains exactly the same elements as $\mathbf{A}[1, n]$ from the original input array. Therefore the original input array has been sorted.

Proof by Contrapositive

Proof by contrapositive is a proof technique for conditional statements. That is, statements of the form “If P , then Q .”

Proof by Contrapositive

Proof by contrapositive is a proof technique for conditional statements. That is, statements of the form “If P , then Q .”

It relies on the fact that $P \Rightarrow Q$ is logically equivalent to $\neg Q \Rightarrow \neg P$.

Proof by Contrapositive

Proof by contrapositive is a proof technique for conditional statements. That is, statements of the form “If P , then Q .”

It relies on the fact that $P \Rightarrow Q$ is logically equivalent to $\neg Q \Rightarrow \neg P$.

To prove $P \Rightarrow Q$ by contrapositive, we show that if the negation of the conclusion is true ($\neg Q$), then the negation of the hypothesis is true ($\neg P$).

Correctness of (one iteration of) Insertion Sort - Setup

Example: one iteration of insertion sort.

- ▶ "In a single iteration, element $x = A[i]$ of the input array is moved to the left until it is no longer smaller than the element to its left; therefore at the beginning of the next iteration, $A[1, i]$ is sorted and contains exactly the same elements as $A[1, i]$ from the original input array."
- ▶ **Claim:** if $x = A[i]$ is moved left until it is larger than the element to its left, then at the end of the iteration $A[1, i]$ is sorted.

Algorithm Insertion Sort

Input: array A of length n

```
1: for  $i \leftarrow 2$  to  $n$  do
2:    $j \leftarrow i$ 
3:   while  $j > 1$  and  $A[j] < A[j - 1]$ 
     do
4:     Swap  $A[j]$  and  $A[j - 1]$ 
5:      $j \leftarrow j - 1$ 
6:   end while
7: end for
```

Correctness of (one iteration of) Insertion Sort - Setup

Example: one iteration of insertion sort.

- ▶ "In a single iteration, element $x = A[i]$ of the input array is moved to the left until it is no longer smaller than the element to its left; therefore at the beginning of the next iteration, $A[1, i]$ is sorted and contains exactly the same elements as $A[1, i]$ from the original input array."
- ▶ **Claim:** if $x = A[i]$ is moved left until it is larger than the element to its left, then at the end of the iteration $A[1, i]$ is sorted.
- ▶ $P = "x = A[i] \text{ is moved left until it is larger than the element to its left}."$
- ▶ $Q = "At the end of the iteration $A[1, i]$ is sorted."$
- ▶ We want to prove $P \Rightarrow Q$.

Algorithm Insertion Sort

Input: array A of length n

```
1: for  $i \leftarrow 2$  to  $n$  do
2:    $j \leftarrow i$ 
3:   while  $j > 1$  and  $A[j] < A[j - 1]$ 
     do
4:     Swap  $A[j]$  and  $A[j - 1]$ 
5:      $j \leftarrow j - 1$ 
6:   end while
7: end for
```

Correctness of (one iteration of) Insertion Sort - Setup

- ▶ $P = "x = A[i] \text{ is moved left until it is larger than the element to its left.}"$
- ▶ $Q = "At \text{ the end of the iteration } A[1, i] \text{ is sorted.}"$
- ▶ We want to prove $P \Rightarrow Q$.

Algorithm Insertion Sort

Input: array A of length n

```
1: for  $i \leftarrow 2$  to  $n$  do
2:    $j \leftarrow i$ 
3:   while  $j > 1$  and  $A[j] < A[j - 1]$ 
     do
4:     Swap  $A[j]$  and  $A[j - 1]$ 
5:      $j \leftarrow j - 1$ 
6:   end while
7: end for
```

Correctness of (one iteration of) Insertion Sort - Setup

- ▶ $P = "x = A[i] \text{ is moved left until it is larger than the element to its left.}"$
- ▶ $Q = "At \text{ the end of the iteration } A[1, i] \text{ is sorted.}"$
- ▶ We want to prove $P \Rightarrow Q$.
- ▶ **Contrapositive** ($\neg Q \Rightarrow \neg P$): If $A[1, i]$ is *not* sorted at the end of the iteration, then $x = A[i]$ is *smaller* than the element to its left.

Algorithm Insertion Sort

Input: array A of length n

```
1: for  $i \leftarrow 2$  to  $n$  do
2:    $j \leftarrow i$ 
3:   while  $j > 1$  and  $A[j] < A[j - 1]$ 
     do
4:     Swap  $A[j]$  and  $A[j - 1]$ 
5:      $j \leftarrow j - 1$ 
6:   end while
7: end for
```

Correctness of (one iteration of) Insertion Sort - Proof

Contrapositive ($\neg Q \Rightarrow \neg P$): If $A[1, i]$ is *not* sorted at the end of the iteration, then $x = A[i]$ is *smaller* than the element to its left.

- ▶ Suppose that at end of iteration i , x was moved to location j .

Algorithm Insertion Sort

Input: array A of length n

```
1: for  $i \leftarrow 2$  to  $n$  do
2:    $j \leftarrow i$ 
3:   while  $j > 1$  and  $A[j] < A[j - 1]$ 
     do
4:     Swap  $A[j]$  and  $A[j - 1]$ 
5:      $j \leftarrow j - 1$ 
6:   end while
7: end for
```

Correctness of (one iteration of) Insertion Sort - Proof

Contrapositive ($\neg Q \Rightarrow \neg P$): If $A[1, i]$ is *not* sorted at the end of the iteration, then $x = A[i]$ is *smaller* than the element to its left.

- ▶ Suppose that at end of iteration i , x was moved to location j .
- ▶ Then since insertion sort swaps x until larger than element to its left, know $x < A[k]$ after iteration i for all $k \in \{j+1, \dots, i\}$.

Algorithm Insertion Sort

Input: array A of length n

```
1: for  $i \leftarrow 2$  to  $n$  do
2:    $j \leftarrow i$ 
3:   while  $j > 1$  and  $A[j] < A[j-1]$ 
     do
4:     Swap  $A[j]$  and  $A[j-1]$ 
5:      $j \leftarrow j-1$ 
6:   end while
7: end for
```

Correctness of (one iteration of) Insertion Sort - Proof

Contrapositive ($\neg Q \Rightarrow \neg P$): If $A[1, i]$ is *not* sorted at the end of the iteration, then $x = A[i]$ is *smaller* than the element to its left.

- ▶ Suppose that at end of iteration i , x was moved to location j .
- ▶ Then since insertion sort swaps x until larger than element to its left, know $x < A[k]$ after iteration i for all $k \in \{j+1, \dots, i\}$.
- ▶ So if $A[1, i]$ not sorted by end of iteration, then must be $A[k]$ with $k < j$ and $A[k] > x$.

Algorithm Insertion Sort

Input: array A of length n

```
1: for  $i \leftarrow 2$  to  $n$  do
2:    $j \leftarrow i$ 
3:   while  $j > 1$  and  $A[j] < A[j-1]$ 
     do
4:     Swap  $A[j]$  and  $A[j-1]$ 
5:      $j \leftarrow j-1$ 
6:   end while
7: end for
```

Correctness of (one iteration of) Insertion Sort - Proof

Contrapositive ($\neg Q \Rightarrow \neg P$): If $A[1, i]$ is *not* sorted at the end of the iteration, then $x = A[i]$ is *smaller* than the element to its left.

- ▶ Suppose that at end of iteration i , x was moved to location j .
- ▶ Then since insertion sort swaps x until larger than element to its left, know $x < A[k]$ after iteration i for all $k \in \{j+1, \dots, i\}$.
- ▶ So if $A[1, i]$ not sorted by end of iteration, then must be $A[k]$ with $k < j$ and $A[k] > x$.
- ▶ Since $A[1, i-1]$ was sorted before iteration (induction), this means that $A[j-1] > A[k] > x$.

Algorithm Insertion Sort

Input: array A of length n

```
1: for  $i \leftarrow 2$  to  $n$  do
2:    $j \leftarrow i$ 
3:   while  $j > 1$  and  $A[j] < A[j-1]$ 
     do
4:     Swap  $A[j]$  and  $A[j-1]$ 
5:      $j \leftarrow j-1$ 
6:   end while
7: end for
```

Correctness of (one iteration of) Insertion Sort - Proof

Contrapositive ($\neg Q \Rightarrow \neg P$): If $A[1, i]$ is *not* sorted at the end of the iteration, then $x = A[i]$ is *smaller* than the element to its left.

- ▶ Suppose that at end of iteration i , x was moved to location j .
- ▶ Then since insertion sort swaps x until larger than element to its left, know $x < A[k]$ after iteration i for all $k \in \{j+1, \dots, i\}$.
- ▶ So if $A[1, i]$ not sorted by end of iteration, then must be $A[k]$ with $k < j$ and $A[k] > x$.
- ▶ Since $A[1, i-1]$ was sorted before iteration (induction), this means that $A[j-1] > A[k] > x$.
- ▶ So x is smaller than the element to its left.

Algorithm Insertion Sort

Input: array A of length n

```
1: for  $i \leftarrow 2$  to  $n$  do
2:    $j \leftarrow i$ 
3:   while  $j > 1$  and  $A[j] < A[j-1]$ 
     do
4:     Swap  $A[j]$  and  $A[j-1]$ 
5:      $j \leftarrow j-1$ 
6:   end while
7: end for
```

Master Theorem

$$T(n) = aT(n/b) + cn^k \qquad T(1) = c$$

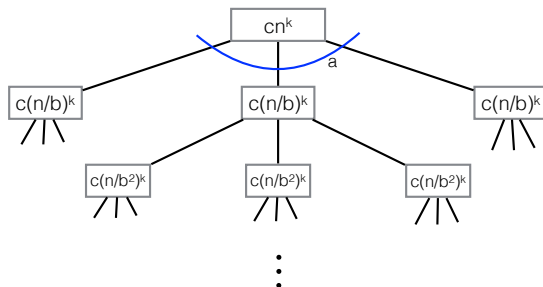
a, b, c, k constants with **$a \geq 1$** , **$b > 1$** , **$c > 0$** , and **$k \geq 0$**

Master Theorem

$$T(n) = aT(n/b) + cn^k$$

$$T(1) = c$$

a, b, c, k constants with $a \geq 1$, $b > 1$, $c > 0$, and $k \geq 0$

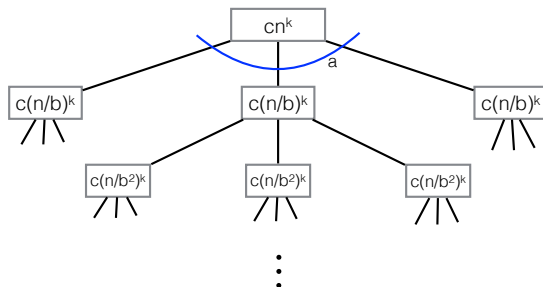


Master Theorem

$$T(n) = aT(n/b) + cn^k$$

$$T(1) = c$$

a, b, c, k constants with $a \geq 1$, $b > 1$, $c > 0$, and $k \geq 0$



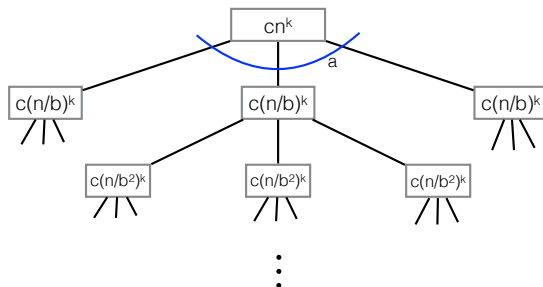
levels: $\log_b n + 1$

Master Theorem

$$T(n) = aT(n/b) + cn^k$$

$$T(1) = c$$

a, b, c, k constants with $a \geq 1$, $b > 1$, $c > 0$, and $k \geq 0$



levels: $\log_b n + 1$

Level i : $a^{i-1}c(n/b^{i-1})^k = cn^k(a/b^k)^{i-1}$

Master Theorem II

Let $\alpha = (a/b^k)$

$$\implies T(n) = cn^k \sum_{i=1}^{\log_b n+1} (a/b^k)^{i-1} = cn^k \sum_{i=1}^{\log_b n+1} \alpha^{i-1}$$

Master Theorem II

Let $\alpha = (a/b^k)$

$$\implies T(n) = cn^k \sum_{i=1}^{\log_b n+1} (a/b^k)^{i-1} = cn^k \sum_{i=1}^{\log_b n+1} \alpha^{i-1}$$

- ▶ Case 1: $\alpha = 1$. All levels the same. $T(n) = cn^k \sum_{i=1}^{\log_b n+1} 1 = \Theta(n^k \log n)$

Master Theorem II

Let $\alpha = (a/b^k)$

$$\implies T(n) = cn^k \sum_{i=1}^{\log_b n+1} (a/b^k)^{i-1} = cn^k \sum_{i=1}^{\log_b n+1} \alpha^{i-1}$$

- ▶ Case 1: $\alpha = 1$. All levels the same. $T(n) = cn^k \sum_{i=1}^{\log_b n+1} 1 = \Theta(n^k \log n)$
- ▶ Case 2: $\alpha < 1$. Dominated by top level.

Master Theorem II

Let $\alpha = (a/b^k)$

$$\implies T(n) = cn^k \sum_{i=1}^{\log_b n+1} (a/b^k)^{i-1} = cn^k \sum_{i=1}^{\log_b n+1} \alpha^{i-1}$$

► Case 1: $\alpha = 1$. All levels the same. $T(n) = cn^k \sum_{i=1}^{\log_b n+1} 1 = \Theta(n^k \log n)$

► Case 2: $\alpha < 1$. Dominated by top level.

$$\implies \sum_{i=1}^{\log_b n+1} \alpha^{i-1} \leq \sum_{i=1}^{\infty} \alpha^{i-1} = \frac{1}{1-\alpha}.$$

$$\implies T(n) = O(n^k)$$

Master Theorem II

Let $\alpha = (a/b^k)$

$$\implies T(n) = cn^k \sum_{i=1}^{\log_b n+1} (a/b^k)^{i-1} = cn^k \sum_{i=1}^{\log_b n+1} \alpha^{i-1}$$

► Case 1: $\alpha = 1$. All levels the same. $T(n) = cn^k \sum_{i=1}^{\log_b n+1} 1 = \Theta(n^k \log n)$

► Case 2: $\alpha < 1$. Dominated by top level.

$$\implies \sum_{i=1}^{\log_b n+1} \alpha^{i-1} \leq \sum_{i=1}^{\infty} \alpha^{i-1} = \frac{1}{1-\alpha}.$$

$$\implies T(n) = O(n^k)$$

$$T(n) \geq cn^k \implies T(n) = \Omega(n^k)$$

Master Theorem II

Let $\alpha = (a/b^k)$

$$\implies T(n) = cn^k \sum_{i=1}^{\log_b n+1} (a/b^k)^{i-1} = cn^k \sum_{i=1}^{\log_b n+1} \alpha^{i-1}$$

► Case 1: $\alpha = 1$. All levels the same. $T(n) = cn^k \sum_{i=1}^{\log_b n+1} 1 = \Theta(n^k \log n)$

► Case 2: $\alpha < 1$. Dominated by top level.

$$\implies \sum_{i=1}^{\log_b n+1} \alpha^{i-1} \leq \sum_{i=1}^{\infty} \alpha^{i-1} = \frac{1}{1-\alpha}.$$

$$\implies T(n) = O(n^k)$$

$$T(n) \geq cn^k \implies T(n) = \Omega(n^k) \implies T(n) = \Theta(n^k)$$

Master Theorem II

Let $\alpha = (a/b^k)$

$$\implies T(n) = cn^k \sum_{i=1}^{\log_b n+1} (a/b^k)^{i-1} = cn^k \sum_{i=1}^{\log_b n+1} \alpha^{i-1}$$

► Case 1: $\alpha = 1$. All levels the same. $T(n) = cn^k \sum_{i=1}^{\log_b n+1} 1 = \Theta(n^k \log n)$

► Case 2: $\alpha < 1$. Dominated by top level.

$$\implies \sum_{i=1}^{\log_b n+1} \alpha^{i-1} \leq \sum_{i=1}^{\infty} \alpha^{i-1} = \frac{1}{1-\alpha}.$$

$$\implies T(n) = O(n^k)$$

$$T(n) \geq cn^k \implies T(n) = \Omega(n^k) \implies T(n) = \Theta(n^k)$$

► Case 3: $\alpha > 1$. Dominated by bottom level

Master Theorem II

Let $\alpha = (a/b^k)$

$$\implies T(n) = cn^k \sum_{i=1}^{\log_b n+1} (a/b^k)^{i-1} = cn^k \sum_{i=1}^{\log_b n+1} \alpha^{i-1}$$

► Case 1: $\alpha = 1$. All levels the same. $T(n) = cn^k \sum_{i=1}^{\log_b n+1} 1 = \Theta(n^k \log n)$

► Case 2: $\alpha < 1$. Dominated by top level.

$$\implies \sum_{i=1}^{\log_b n+1} \alpha^{i-1} \leq \sum_{i=1}^{\infty} \alpha^{i-1} = \frac{1}{1-\alpha}.$$

$$\implies T(n) = O(n^k)$$

$$T(n) \geq cn^k \implies T(n) = \Omega(n^k) \implies T(n) = \Theta(n^k)$$

► Case 3: $\alpha > 1$. Dominated by bottom level

$$\begin{aligned} \implies \sum_{i=1}^{\log_b n+1} \alpha^{i-1} &= \alpha^{\log_b n} \sum_{i=1}^{\log_b n+1} \left(\frac{1}{\alpha}\right)^{i-1} \leq \alpha^{\log_b n} \frac{1}{1-(1/\alpha)} \\ &= O(\alpha^{\log_b n}) \end{aligned}$$

Master Theorem II

Let $\alpha = (a/b^k)$

$$\implies T(n) = cn^k \sum_{i=1}^{\log_b n+1} (a/b^k)^{i-1} = cn^k \sum_{i=1}^{\log_b n+1} \alpha^{i-1}$$

► Case 1: $\alpha = 1$. All levels the same. $T(n) = cn^k \sum_{i=1}^{\log_b n+1} 1 = \Theta(n^k \log n)$

► Case 2: $\alpha < 1$. Dominated by top level.

$$\implies \sum_{i=1}^{\log_b n+1} \alpha^{i-1} \leq \sum_{i=1}^{\infty} \alpha^{i-1} = \frac{1}{1-\alpha}.$$

$$\implies T(n) = O(n^k)$$

$$T(n) \geq cn^k \implies T(n) = \Omega(n^k) \implies T(n) = \Theta(n^k)$$

► Case 3: $\alpha > 1$. Dominated by bottom level

$$\implies \sum_{i=1}^{\log_b n+1} \alpha^{i-1} = \alpha^{\log_b n} \sum_{i=1}^{\log_b n+1} \left(\frac{1}{\alpha}\right)^{i-1} \leq \alpha^{\log_b n} \frac{1}{1-(1/\alpha)}$$

$$= O(\alpha^{\log_b n})$$

$$\implies T(n) = \Theta(n^k \alpha^{\log_b n}) = \Theta(n^k (a/b^k)^{\log_b n}) = \Theta(a^{\log_b n})$$

$$= \Theta(n^{\log_b a})$$

Master Theorem III

Theorem (“Master Theorem”)

The recurrence

$$T(n) = aT(n/b) + cn^k$$

$$T(1) = c$$

where a, b, c , and k are constants with $a \geq 1$, $b > 1$, $c > 0$, and $k \geq 0$, is equal to

$$T(n) = \Theta(n^k) \text{ if } a < b^k,$$

$$T(n) = \Theta(n^k \log n) \text{ if } a = b^k,$$

$$T(n) = \Theta(n^{\log_b a}) \text{ if } a > b^k.$$